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WHAT THE LATENT TRAJECTORY OF A RECURRENT-DEPTH TRANSFORMER ENCODES, AND WHAT OUR MEASUREMENTS WERE ACTUALLY MEASURING

Abstract. We study the latent trajectory of Huginn-3.5B, a recurrent-depth transformer that applies one weight-tied block a variable number of times before decoding. Two lines of results are reported. First, the recurrent map has a sharp qualitative structure: a single instruction word switches it between a settling regime and a damped period-6 rotation (36/36 paired cells at an identical token count), the switch is causally controlled by one row of the embedding matrix, the regime is a property of the prompt rather than of the token position sampled, and swapping the map's injected parameter mid-trajectory re-aims the dynamics with no hysteresis in 62 of 64 switches. Second, and less comfortably, a large part of what this project measured as capability was answer formatting: first-token scoring reads 0.125 where the answer is actually present in 0.781 of generations, depth raises prose-opening twentyfold while leaving the rate of leading with the answer flat, and supplying the output format recovers the model's published benchmark number by two independent routes. We report the retractions this produced alongside the results, since several of them were caused by our own instruments rather than by the model. This is a draft report; several stated limits are open.

§1. Introduction

A recurrent-depth transformer reuses one weight-tied block many times at inference, so the sequence of latent states it passes through is often read as a trace of reasoning — something whose shape should reflect the computation being performed. We tested that reading on Huginn-3.5B [1] and report two things.

The first is that the map has real, sharp, causally controllable structure, but that this structure is selected by the surface form of the instruction

Key words and phrases: recurrent-depth transformers and latent reasoning and interpretability and evaluation protocols and dynamical systems.

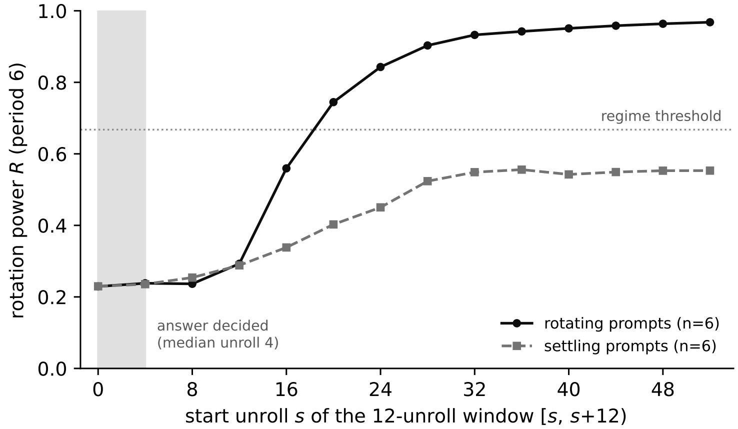

Figure 1. Rotation power in a sliding 12-unroll window, median over six rotating and six settling prompts. The axis is the window start: each point summarises unrolls $[s,\, s{+}12)$, twelve being the smallest window in which a period-6 orbit is defined at all. The two families are indistinguishable until the window beginning at unroll 16, by which point the answer has long been settled (shaded; median unroll 4). Higher R means more of the trajectory's power sits at period 6.

rather than by the task. The second is that a substantial part of our own measured capability record — and, we argue, of the standard way such models are scored — was measuring whether the model emits its answer in the position a scorer reads, not whether it has the answer.

We report both, together with the claims we withdrew in the process. The withdrawals are included deliberately: in several cases the instrument, not the model, produced the result, and a report that omitted them would misrepresent how reliable the remainder is.

§2. Model and Protocol

Huginn-3.5B (tomg-group-umd/huginn-0125, revision bb6621b6...) has the structure

$$\begin{aligned} &\text{embed} \to \text{prelude}(2) \to \big[\,\text{core\_block}(4)\,\big]^{\times r} \\ &\qquad \to \text{coda}(2) \to \text{ln\_f} \to \text{lm\_head}, \end{aligned}$$

where the prelude output e is re-injected at every unroll through the adapter, which receives the concatenation of the current state and e. Following the DEQ reading [6], we treat e as the map's parameter and the latent state h as its state; the released source supports this, since the block index is threaded through the unroll but never enters the block computation.

All measurements are inference-only in float32 and hook the output of core_block[-1]. Unless stated, depth is $r = 32$ or $r = 48$. Every forward is seeded, because the initial state is drawn from an unseeded generator by default.

§3. The Recurrent Map Has a Sharp, Causally Controlled Regime

3.1. One instruction word switches the regime. Holding the sequence, the required answer and the token count fixed at 53, the instruction "report the largest symbol" produces a damped period-6 rotation in 36 of 36 paired cells, while "largest element" and "largest item" produce settling in 0 of 36. The selecting property is not semantic, not orthographic, and not tokenisation: the semantic neighbours of symbol rotate in 0 of 144 orbits, those of element in 96 of 120, and form-matched controls chosen to denote nothing similar (symptom, cymbal, symmetry) rotate in 72 of 72.

A single threshold on a continuous rotation statistic reproduces the binary label on 691 of 696 orbits, and on 688 of 696 held out leave-one-bank-out.

3.2. The switch is causally controlled by one embedding row. Editing a single row of the embedding matrix, with the prompt string and every token identifier untouched, flips the regime. Both instrument gates are bit-exact: substituting the row for itself reproduces the unpatched orbit at 0.000e+00, and substituting the target's row reproduces the target's own orbit.

Scope. The within-regime control failed in that experiment and was only partly resolved afterwards: rescaling the interpolated row to constant norm rescues one noun pair and not another. The licensed statement is therefore that the embedding row causally determines the regime within a small, direction-dependent neighbourhood — not that the regime is a simple function of the embedding.

3.3. The regime belongs to the prompt, not to the position measured. Every measurement above reads one token position. Because the published figures for this model are drawn at interior positions, we re-measured the rotation statistic at every position. Prompts we label rotating rotate at 62.0% of their positions (range 0.614–0.623 over nine prompts); prompts we label settling rotate at 0.000 — no overlap and no intermediate case. Within a rotating prompt, rotation switches on at a single boundary and holds to the end.

3.4. No hysteresis, and the contraction rate predicts a timescale. Swapping e mid-trajectory and leaving it swapped, 62 of 64 switches take, at switch points from unroll 1 to unroll 48: the regime follows the parameter the map currently has rather than the trajectory's history. Entering rotation takes a median of 15.5 unrolls and leaving it 1.0 — a fifteenfold asymmetry measured by the identical detector. The entering time brackets a prediction registered before the run from the measured contraction rate $\rho \approx 0.83$, namely $\ln(0.05)/\ln(0.83) = 16.1$ unrolls.

Scope. The detector uses a sliding window that carries its own lag, so the measured relaxation is an upper bound. The asymmetry, measured identically in both directions, is the robust part.

§4. What the Capability Measurements Were Measuring

4.1. The answer is present but not in the scored position. Generating and reading the text, rather than reading the rank of the gold token at the first generated position, changes the picture. First-token exact match reads 0.125 where the answer is actually produced in 0.781 of generations, against a measured cross-item false-positive rate of 0.042–0.152. The token that displaces the gold is The: it opens 39.4% of 1260 banked generations. On tasks the model can do, the gold's rank at the final unroll never exceeds 35 across 2720 draws.

4.2. Depth increases prose, not answers. Across $r = 2, 4, 8, 16, 32$ the rate at which a generation opens with The runs $0.032 \to 0.099 \to 0.631 \to 0.560 \to 0.647$, while the rate at which it opens with the gold does not move ($p = 0.515$). More depth therefore makes the standard first-token metric worse while the underlying output improves.

4.3. Supplying the format recovers the published number, by two routes. The published ARC-Easy figure for this model is 0.699 at $r = 32$. Reading the released evaluation code pins the protocol to zero-shot lm-eval-harness with normalised accuracy; that is our bare-prompt, no-option-list, character-normalised arm, which reads 0.658. Independently, five in-context examples under option-letter argmax give 0.723, paired, with 35 items fixed against 4 broken (exact McNemar $p = 3.35\text{e}{-07}$).

4.4. Instruction fails where demonstration succeeds. Instructing the model to "reply with only the answer" yields no improvement over a bare prompt; asking it to place its answer in \boxed{} produced fewer such answers than not asking (2 of 36 against 4 of 36). Two in-context examples, by contrast, move oracle accuracy by $+0.221$ (exact McNemar $p = 9.3\text{e}{-09}$) and take three families from 0.000 to 1.000 at the final unroll.

Placing the identical instruction in the system turn rather than the user turn—the form another group used to suppress prose in this model—buys one item in twenty-four (paired, $p = 1.0$), so the failure is not a property of the turn. The one qualified exception is an instruction that does not fight the prose frame but appends to it: asking the model to work through the problem and then write Answer: on the last line raises parsed-exact accuracy from 0.000 to 0.167, gaining four items and losing none ($p = 0.125$, $n = 24$), with literal outputs such as 'Answer: 1'. Suggestive rather than established, and consistent with the prose being where the computation happens.

4.5. Context costs depth rather than accuracy. Padding a bare prompt with irrelevant prose to the five-shot token length (length match verified, median ratio 1.000) collapses the gold's starting rank from 31 to 329, where five real exemplars leave it at 58.5. Oracle accuracy barely moves (0.500 to 0.479) while the depth at which the answer becomes available rises from 4.73 to 7.29 unrolls ($p = 0.0001$).

4.6. The generated prose is computation. Measuring the trajectory at every generated position rather than only the first, the forward pass that carries the answer is deeper than position 0 ($4.07 \rightarrow 7.89$, $p = 0.0031$) and deeper than the surrounding prose positions (7.89 against 3.65, $p =$

0.0004). The effect concentrates on arithmetic and is near-absent on copying.

§5. The Regime and the Answer Occupy Different Windows

The two preceding sections are connected by a timing fact that reframes both. The rotation statistic is computed over the tail of the trajectory; the answer, by contrast, is best-ranked at a median unroll of about four. Measuring the same statistic over the decision window (unrolls 0–11, the earliest a period-6 orbit is definable at all, since two cycles are required) and over the tail separates two facts that had been conflated. Over the decision window, rotating and settling prompts are indistinguishable: 0.229 against 0.230, a separation of 0.000. Over the tail the same prompts separate completely, 0.862 against 0.298. A sliding twelve-unroll window dates the divergence (Figure 1). Indexing each window by its start, the two curves are identical through the window beginning at unroll 12 (0.29 against 0.29) and first separate at the window beginning at unroll 16 (0.56 against 0.34); the rotating curve then crosses the regime threshold and saturates at 0.96, while the settling curve flattens at 0.55 and never reaches it.

The early window is not quiet. Its non-DC power exceeds the tail's by about two orders of magnitude (1.6×10^5 against 1943 and 59); it is energetic and simply carries no period-6 structure.

This changes what our own behavioural null means. We had found that a causally induced regime change alters almost nothing in what the model answers—one of eighteen paired units. That is not evidence that the regime is a dynamical epiphenomenon: the manipulation acted on a property that had not yet come into existence when the readout was settled. This is a timing claim rather than a causal one. It does not show that the regime could not affect behaviour, only that on this architecture the answer is fixed roughly twelve unrolls before the regime becomes measurable. It also bounds the instrument: no period-6 statistic can describe the state at the moment of decision, and we state this wherever the regime is connected to behaviour.

§6. The Three Starting Hypotheses

H1 (settle / loop / drift). Answered, in the form "both, at different levels". Unbounded drift is excluded by construction, since every recorded state is a normalisation output on a sphere. A large cycle exists across the four core blocks; within a block the state settles or rotates according to the instruction token.

H2 (harder problems recruit more depth). Architecturally untestable on this model. The hypothesis as the literature states it concerns per-position adaptive depth, which adaptive-computation models measure directly and confirm: ponder time grows with difficulty when a halting cost is trained in [2], a universal transformer with dynamic halting reports mean per-symbol depth rising $2.3 \pm 0.8 \to 3.1 \pm 1.1 \to 3.8 \pm 2.2$ as the number of required supporting facts goes from one to three [4], and the objective is stated directly as computation that grows with problem complexity rather than with input size [3]. Huginn has no halting mechanism: the whole sequence is unrolled together, so every position receives the same r and there is nothing to allocate. Depth does buy capability at fixed difficulty, but difficulty within a task moves recruited depth by $+0.23$ unrolls against task identity's $+5.56$. A difficulty axis often proposed for recurrence, sequential state tracking, is moreover out of reach for this architecture class on theoretical grounds: parity is the S_2 word problem and is argued to be unsolvable by these models, with many state-tracking problems harder than S_5 [5].

H3 (contraction implies no running register). Its strong form overstates its evidence. Contraction is real, but the limit set has ample room to separate states, and the running count is linearly decodable from the latents.

§7. Reliability: Controls, Retractions, and Instrument Nulls

Several results in this project were produced by instruments rather than by the model. We record them because the reliability of the remainder depends on it.

- A headline behavioural null was vacuous: the correctness variable it compared was false in 432 of 432 orbits, so there was no variance for the intervention to change. Re-tested on an outcome that does vary, the original conclusion returned.
- A coordinate-geometry headline was withdrawn by its own registered control within hours of being written.
- A linear probe of the class used for several of our null results cannot recover a label that is a guaranteed deterministic function of its own input features (0.690 against a 0.600 baseline, where the correct one-dimensional statistic reaches 0.980). Those nulls are uninformative rather than negative.

- The extraction rule dominates the number: over 1424 banked generations, exact match recovers 0.038 of the answers and a last-number rule 0.240 at half the false-positive rate.

§8. Known Limitations

We do not reproduce the orbit reported for this model on a prompt of the kind its authors use to illustrate it. Sweeping every period the tail can resolve, rather than the period-6 bin alone, finds no peak anywhere: the median non-DC power of that trajectory is 2.096, against 665–3139 for prompts we call rotating, and two further free-form prompts read 0.023 and 0.363. Our detector agrees with its own spectral definition to 4×10^{-8}, and our rotating prompts peak at period 6 specifically rather than merely somewhere. We therefore state the negative narrowly: numerical-reasoning content is not sufficient to produce the orbit we measure. We ran a prompt in the shape of their example rather than their exact string, and their figures are projections of a 128-iteration run against our 64.

- Our benchmark numbers and the published figure share an uncorrected surface-form (first-token) bias inherent to likelihood scoring.
- We do not reproduce the published orbit on the paper's own word-problem example (0 of 35 positions). The leading explanation is that our detector is tuned to a single period; a period sweep was running at the time of writing.
- The published attribution of orbiting to "prompts requiring numerical reasoning" is contradicted by our data on three independent grounds, while the phenomenon itself is theirs.
- Comparisons to two related works are cross-checkpoint or cross-protocol and are not directly numerically comparable.

§9. Conclusion

The recurrent map of Huginn-3.5B carries a sharp, causally controllable qualitative structure that is selected by the surface form of the instruction rather than by the task it states, and that has no measurable consequence for the answer under the tests we could run. Separately, a substantial part of what we and the standard protocols measured as capability was the model's willingness to emit its answer where a scorer looks. Supplying the format, by demonstration or by matching the evaluation protocol, recovers the published benchmark number.

§10. Contribution

(Placeholder — to be completed before submission. One entry per team member, describing that member's specific contribution.)

- Author One — placeholder: e.g. experiment design, kernel implementation, analysis of the regime results (Section 3).
- Author Two — placeholder: e.g. evaluation-protocol experiments and analysis (Section 4), literature comparison.
- Mentor: Serguei Barannikov — placeholder: problem formulation, hypotheses H1–H3, supervision.

Acknowledgments

(Placeholder.)

Appendix §A. Numerical Summary

Quantity	Value
Regime switch by one instruction word	36/36 rotating vs 0/36, token count fixed at 53
Semantic control	neighbours of symbol 0/144; of element 96/120
Form-matched control	72/72 rotating
Threshold reproduces binary label	691/696; 688/696 held out
Embedding-row edit flips regime	gates bit-exact at 0.000e+00
Regime by position	rotating prompts 62.0% of positions; settling 0.000
Mid-trajectory parameter swap	62/64 switches take
Relaxation asymmetry	entering 15.5 unrolls, leaving 1.0
Predicted from $\rho \approx 0.83$	16.1 unrolls
First-token vs produced answer	0.125 against 0.781
Chance rate for containment	0.042–0.152
The as opening token	39.4% of 1260 generations
Gold's worst final rank (feasible tasks)	35, over 2720 draws
ARC-Easy, published	0.699 at $r = 32$
ARC-Easy, matched protocol	0.658
ARC-Easy, five-shot letter argmax	0.723 ($p = 3.35\text{e}{-07}$)
Two-shot effect on oracle accuracy	+0.221 ($p = 9.3\text{e}{-09}$)
Irrelevant padding, start rank	31 $\rightarrow$ 329 ($p = 2.9\text{e}{-09}$)
Answer position vs prose positions	7.89 against 3.65 unrolls ($p = 0.0004$)

Table 1. Principal quantitative results. Inference-only on the pinned revision.

Appendix §B. Withdrawn and Amended Claims

(Placeholder for the full list. The four summarised in Section 6 are the load-bearing ones.)

Appendix §C. Template and Encoding Notes

The template repository ships zapiski.cls in CP1251 and its README requires the .tex to be CP1251 as well, but the distributed template.tex is UTF-8. That mismatch makes the Russian masthead strings inside the class decode incorrectly, and the distributed template works around it with explicit \renewcommand overrides.

This document instead follows the README: it is written in CP1251 with \usepackage[cp1251]{inputenc}, matching the class. With the encodings aligned the masthead renders correctly and no override is required — verified by compiling a minimal probe without them.

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Что кодирует латентная траектория трансформера с рекуррентной глубиной

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Аннотация

Исследуется латентная траектория модели Huginn-3.5B — трансформера с рекуррентной глубиной. Показано, что одно слово в инструкции переключает рекуррентное отображение между режимом сходимости и затухающим вращением с периодом 6, причём переключение причинно управляется одной строкой матрицы вложений. Показано также, что значительная часть измерений качества модели фактически измеряла формат ответа, а не его наличие. Приводятся опровергнутые и уточнённые утверждения.

Ключевые слова: трансформеры с рекуррентной глубиной, латентные рассуждения, интерпретируемость, протоколы оценки.